D 12039 **(Pages : 2)** **Name.....**

Reg. No.....

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021

Physics/Applied Physics

PHY 3B 03/APH 3B 03—ELECTRODYNAMICS—I

(2019—2020 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. Explain Coulomb's law in electrostatics.
2. Give an expression for the electric field due to a surface charge. Explain the terms involved.
3. How is electric field and potential related in electrostatics ? What is Laplace's equation in electrostatics ?
4. Write down the relation connecting polarization and electric field in linear dielectrics. What do you mean by a polar dielectric ?
5. Write down Gauss's law in presence of a dielectric.
6. Give the relation connecting permittivity and susceptibility in linear dielectrics. Explain the terms involved. How is relative permittivity connected to susceptibility ?
7. Check whether the following statements are true or false : (i) stationary charges create constant magnetic field ; and (ii) a moving point charge cannot constitute a steady current.
8. Give Biot-Savart law. Explain the terms involved.
9. Give the differential and integral forms of Ampere's law in magnetostatics.
10. Write down the relation connecting magnetic flux density B and magnetic vector potential A. What is the divergence of A ?

Turn over

11. Explain the physical meaning of the equation of continuity.

12. What is the basic difference between electric and magnetic polarizations ?

(8 × 3 = 24 marks)

Section B (Paragraph/Problem Type)

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. Prove that the divergence of the curl of a vector is always zero.

14. Check whether $\mathrm{E} = k\left[xy\hat{x} + 2yz\hat{y} + 3xz\hat{z}\right]$, is a possible electrostatic field ? Here, k is a constant with appropriate units.

15. Show that the energy of an ideal dipole p in an electric field E is given by $\mathrm{U} = -p.\mathrm{E}$.

16. Compare electrostatics and magnetostatics in terms of the Maxwell's equations applicable. Compare the nature of the fields with respect to their source.

17. Define the term surface current density. A current I flows down a wire of radius a. If the current is uniformly distributed over the surface, what is the surface current density ?

18. Write an expression for the field of a magnetized object. Explain the terms involved.

19. Explain the domain model of ferromagnets.

(5 × 5 = 25 marks)

Section C (Essay Type)

Answer any **one** *question.*
The question carries 11 *marks.*

20. Discuss the divergence of a vector function and give its geometrical interpretation. Explain Green's theorem and illustrate it geometrically.

21. Explain the term electric potential. Obtain an expression for the electric potential of a localized charge distribution.

(1 × 11 = 11 marks)